Ericsson PE (WHI-PE-14) Hardware/Environmental Fault Resolution SOP

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_3208

Description

The PE (Path Element) role in the Ericsson network architecture is responsible for providing high-speed packet switching and routing capabilities. The WHI-PE-14 node, located at the Whitefield East POP, is a critical component of the network infrastructure. The failure domain in question, ALARM_3208, is a hardware/environmental fault that can occur due to ASIC memory parity errors, line card crashes, or power supply redundancy loss. This SOP outlines the steps to diagnose and resolve this type of failure, which is critical to maintaining network uptime and ensuring high-quality service to customers. The underlying protocol and hardware mechanics that contribute to this failure domain include the use of ASICs (Application-Specific Integrated Circuits) to implement high-speed packet processing, which can be prone to memory parity errors, and the reliance on line cards to provide high-speed interfaces, which can crash due to hardware or software issues. Additionally, the power supply redundancy mechanism, designed to ensure continuous operation in the event of a power supply failure, can be compromised if one of the redundant power supplies fails or is removed.

Pre-requisite Checks

- Verify that the Whitefield East POP is accessible and dispatch remote hands if necessary.
- Check the redundant link status to ensure it is functioning properly.
- Confirm that the node is in a stable state and not experiencing any other alarms or issues.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system redundancy, Expected Output: Redundancy Status: Active, Description: This command verifies the status of the redundancy mechanism, ensuring that the node is functioning in an active redundancy mode.

Command: show system asic, Expected Output: ASIC Status: Normal, Description: This command checks the status of the ASIC, verifying that it is operational and not experiencing any errors.
```

```
Command: show system power supply, Expected Output: Power Supply Status: Redundant, Description: This command checks the status of the power supply, verifying that it is functioning in a redundant mode.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC, Command: replace system asic 1, Description: This step involves replacing the faulty ASIC with a new one, ensuring that the node is restored to a stable state.

Step 2: Step: Replace faulty line card, Command: replace system line-card 1, Description: This step involves replacing the faulty line card with a new one, ensuring that the node is restored to a stable state.

Step 3: Step: Verify power supply redundancy, Command: show system power supply, Description: This step involves verifying the status of the power supply redundancy, ensuring that it is functioning properly.

Post-Check Verification

```
Command: show system status, Expected Output: System Status: Normal, Description: This command verifies the overall system status, ensuring that it is stable and functioning properly.

Command: show system alarms, Expected Output: Alarm Status: None, Description: This command checks the alarm status, verifying that there are no active alarms on the system.
```

Roll-back Procedures

- Step: Restore original configuration, Command: config load default, Description: This step involves restoring the original configuration, ensuring that the node is returned to its previous state.

- Step: Revert changes made during resolution, Command: rollback system changes, Description: This step involves reverting the changes made during the resolution process, ensuring that the node is returned to its previous state.